FACULTY OF COMPUTING

SECJ3533-02 ARTIFICIAL INTELLIGENCE

SEMESTER 5

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ASSIGNMENT 2: STATE SPACE SEARCH

THEME: SMART EDUCATION

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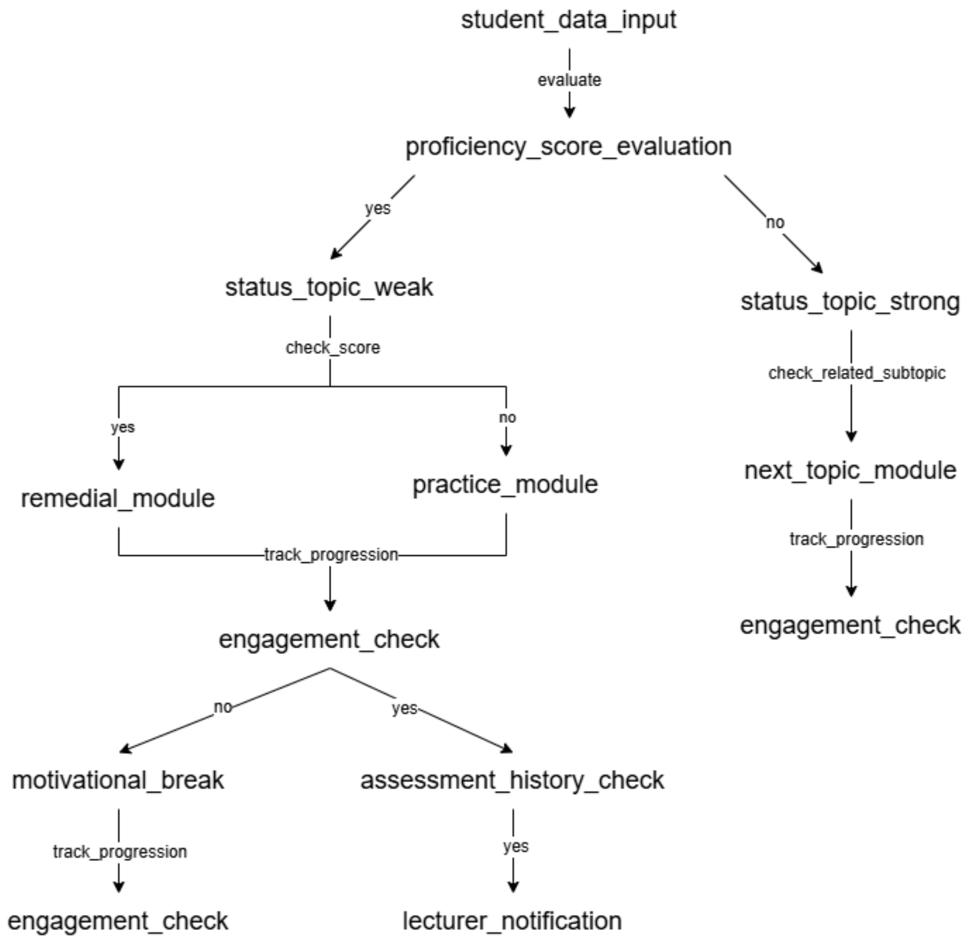
Problem Formulation by State Space Graph

1.0 Smart Education Assistance



1.1 Explanation States and Actions

States	**1. student_data_input** ● The instructor uploads or selects class-related data such as attendance, grades, or assessments. **2. profieciency_score_evaluation**

	● The system processes the class data to evaluate class performance levels. **3. status_topic_weak** ● Indicates that the class or a portion of the class is performing below the required threshold. **4. status_topic_strong** ● Indicates that the class is performing above the expected standard. **5. remedial_module** ● If the students lack prerequisites, the system provides teaching suggestions, remedial strategies, or targeted improvements. **6. practice_module** ● If they have the prerequisites, The system provides extra practice to reinforce the current topic. **7. next_topic_module** ● The system will provide new and more difficult contents for the students that have mastered the current material. **8. engagement_check** ● The system evaluates student engagement (attendance, participation, LMS activity, etc.). **9. motivational_break** ● If engagement is low, the system triggers a “break” like a game or video to prevent burnout. **10. Assessment_history_check**

	● If engagement is high, the system looks at the long-term trend. **11. lecturer_notification** ● Final state where the system sends analytics results, alerts, or recommendations to the instructor.
Actions	**1. analyze** ● Triggers the system to process and analyse class performance data. **2. yes** ● A condition is met (e.g., low performance detected, risk identified). **3. no** ● A condition is not met (e.g., performance above threshold, no risks found). **4. detect_issues** ● The system examines data to find low-performing areas or problematic topics. **5. recommend** ● Generates improvement suggestions or interventions for the instructor. **6. alert** ● Sends warnings or flags regarding at-risk or problematic class segments.

1.2 Problem Formulation and Explanation

Critical	Details	Explanation
Initial State	student_data_input	The initial state of student_data_input implies the system's focus on pulling all the student records and the course structures to begin the evaluation process.
Action	Let S(x) = Set of action-state pairs S(student_data_input) = {evaluate: proficiency_score_evaluation} S(proficiency_score_evaluation) = {yes:status_topic_weak, no: status_topic_strong} S(status_topic_weak) = {check_score: (yes: practice_module, no: remedial_module)} S(remedial_module) = {track_progression: engagement_check} S(practice_module) = {track_progression: engagement_check} S(status_topic_strong) = {check_related_subtopic: next_topic_module} S(next_topic_module) =	The process begins with the algorithm that accesses the student information to assess and categorize them according to their present mastery skills. After that, the system starts checks for a low proficiency student by activating check_prereq. The practice_module will be given to the students that have gone through the necessary prerequisites successfully and have mastered them, whereas the non-masters will continue to use the remedial_module. A student who has a solid grasp of the subject matter will be moved to the next_topic_module. The system will then enter a real-time monitoring loop via the engagement_check. If the engagement is maintained, the system will move to evaluate long-term performance through assessment_history_check. If the system detects that the student is disengaged, the student is assigned as motivational_break before returning to the loop. Finally, if the historical analysis proves a

	{track_progression: engagement_check} S(engagement_check) = {yes: assessment_history_check, no: motivational_break} S(motivational_break) = {track_progression: engagement_check} S(assessment_history_check) = {yes: lecturer_notification}	significant decrease or indicates a knowledge gap of the whole class, the algorithm achieves its objective by causing a lecturer_notification to be sent for the prompt teacher intervention.
Goal	lecturer_notification	The goal that we want to achieve is to notify the lecturer when a student shows disengagement during their learning process. This allows the lecturer to intervene and provide appropriate support to help the student get back on track with their studies.
Path cost	student_data_input, proficiency_score_evaluation, status_topic_weak, practice_module, engagement_check, assessment_history_check, lecturer_notification	The path cost is the execution time taken for the system to reach the goal which is to alert the lecturer about student disengagement based on engagement monitoring and assessment history checks.

1.3 Example solution:

1. student_data_input , proficiency_score_evaluation , status_topic_weak , remedial_module, engagement_check , assessment_history_check , lecturer_notification